

ALL PAIR SHORTEST
PATH (APSP)

Let's revisit

SINGLE-SOURCE SHORTEST PATH (SSSP)

Input: A directed graph $G=(V,E)$,
starting vertex s , a possibly
negative length l_e , for each edge e

(G may contain $-ve$
cycle as well)

Output - ① For all destinations
 $v \in V$, compute the
length of the shortest $s \rightsquigarrow v$
path, or

② Output a $-ve$ cycle

Optimal Substructure

Lemma - Let $G = (V, E)$ be a directed graph with edge lengths l_e and start vertex s .

For any destination $v \in V$ and every $i \in \{1, 2, \dots\}$, let P be the shortest $s \rightsquigarrow v$ path with at most i edges.

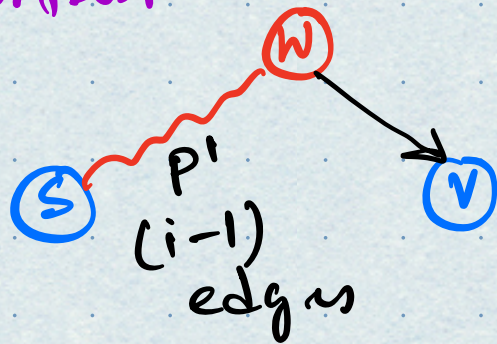
Case - 1 - P has $(i-1)$ or fewer edges

$\Rightarrow$ P is the shortest $s \rightsquigarrow v$ path with at most $i-1$ edges.

Case - 2 - P has i edges.

$\Rightarrow P'$ is the Shortest

$S \rightsquigarrow W$ path with
at most $(i-1)$
edges



THE RECURRENCE

Define $L_{i,v}$ = length of Shortest
 $S \rightsquigarrow V$ path with
 $\leq i$ edges.

with cycles allowed.

defined as $+\infty$ if no
 $S \rightsquigarrow V$ path with $\leq i$
edges.

For every $v \in V$ and every
 $i \in \{0, 1, 2, \dots\}$.

$$L_{i,v} = \min \begin{cases} L_{i-1,v} & \text{Case -1} \\ \min_{(w,v) \in E} L_{i-1,w} + L_{w,v} & \text{Case -2} \end{cases}$$

BELLMAN - FORD

$A = n \times n$ dimensional array
// budget $\times$ destinations

// base cases

$$A[0, s] = 0, \quad A[0, v] = +\infty, \\ \forall v \neq s.$$

Recurrence

for $i = \{1, 2, \dots, n-1\}$

for each $v \in V$

Running time

$$= O(mn)$$

$$1 + \text{in-deg}(v)$$

Solve problems for v

$$A[i, v] = \min \begin{cases} A[i-1, v] \\ \min_{(w, v) \in E} A[i-1, w] + l_{w, v} \end{cases}$$

Running time

$$= O(mn)$$

$$O\left(n \left(\sum_v 1 + \text{in-deg}(v) \right)\right)$$

$$= O(n^2 + mn) = O(mn)$$

What about -ve cycles?

* Detecting -ve cycles

claim: G has no -ve reachable from s if and only if in the extended Bellman-Ford algorithm

$$A[n-1, v] = A[n, v], \forall v \in V.$$

- Add auxiliary vertex to detect -ve cycle.. reachable from s or otherwise.

works?

APSP

QIP :- A directed graph $G=(V,E)$
a possibly -ve length edge (l_e), $\forall e \in E$

No distinguished "Source" vertex.

O/P: ① Compute the length of the
Shortest $u-v$ path, $\forall$ pairs of
vertices $u, v \in V$, or

② O/P a -ve cycle.

Idea - Why not try SSSP
in rounds, $\forall v \in V$.

(Consider each vertex as a
Source vertex)

If non-negative lengths $\Rightarrow$

Run Dijkstra n times

$$\Rightarrow O(m \times n \log n)$$

$$O(n^2 \log n)$$

Sparse

$$O(n^3 \log n)$$

Dense.

Too much Space!

Q: - Is $O(n^{3-\epsilon})$ is possible, for any $\epsilon > 0$!

One of the important open problem in Fine-Grained Complexity!

It feels hard to beat n^3 .

Afterall, #edges / pairs $\binom{n}{2}$
 $O(n^2)$

But,
Maybe some computation is
Simultaneously Relevant to
multiple problems

i.e.,

Recall

Matrix multiplication?

Classical - $O(n^3)$

Strassen - $O(n^{2.807})$

Alman-Williams - $O(n^{2.371})$

Trivial LB - $\Omega(n^2)$

Open - $2 \leq \omega \leq 2.37$

If possibly -ve lengths

$\Rightarrow$ Run Bellman-Ford
n times

$$\Rightarrow O(m \times n^2)$$

$O(n^3)$
Sparse

$O(n^4)$
Dense

only +ve lengths
Run Dijkstra.

Floyd - Warshall Alg.

* $O(n^3)$ for APSP.

✓ * Works even with negative edge lengths.

✓ * At least as good as n invocations of Bellman-Ford, even on sparse graphs.

* For non-negative edge lengths Comp. with n Dijkstra's in dense graphs.

Applications - Finding transitive closure of binary relation

(Game theory) - Schulz voting rule.

Open - $O(n^{3-\epsilon})$ algo. for

APSP in dense graphs.

Optimal Substructure

Q. How to define "Subproblems" on graph problem.

Idea - Order the vertices

$V = \{1, \dots, n\}$ arbitrarily

let $V^{(k)} = \{1, 2, \dots, k\}$

(similar to what we did for SSSP).

Lemma - Suppose graph G has no negative cycle.

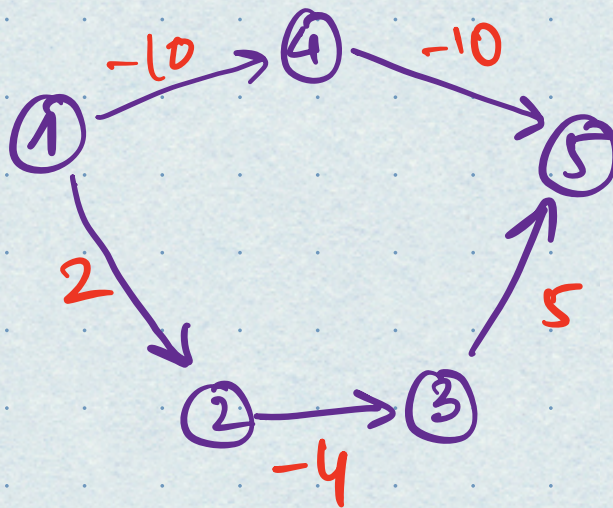
Fix Source $i \in V$,
destination $j \in V$ & prefix length k

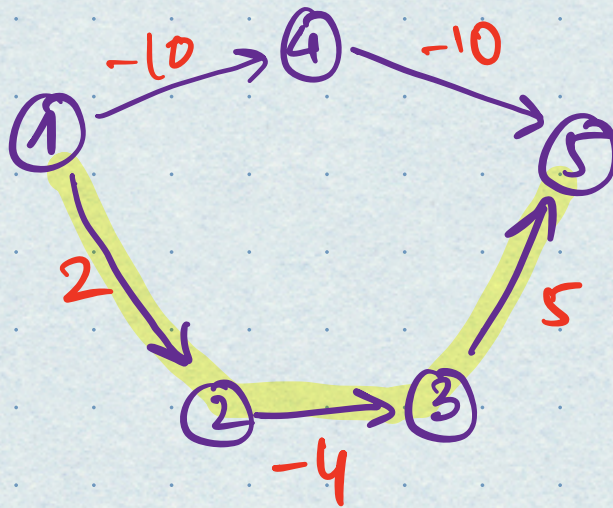
$P =$ Shortest (cycle-free) $i \rightsquigarrow j$
path with all **internal nodes**
in $V^{(k)}$.

Ex. $i=1, j=5$

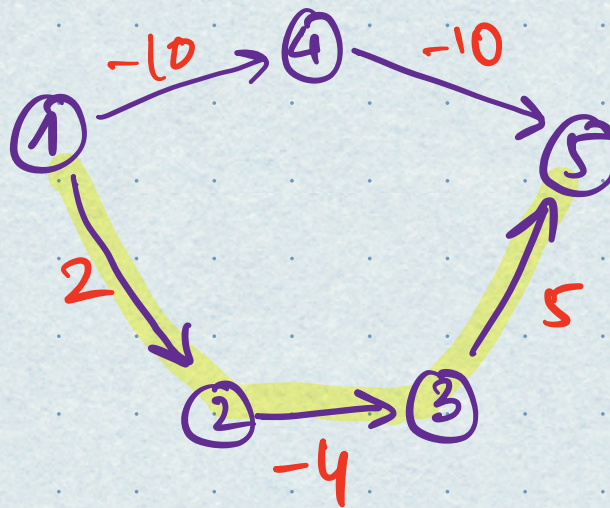
$k = \{0, 1, 2\}$

No path.





$k = 3$: 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 5



$k = 4$: 1 $\rightarrow$ 4 $\rightarrow$ 5

Lemma - Suppose graph G has no negative cycle.

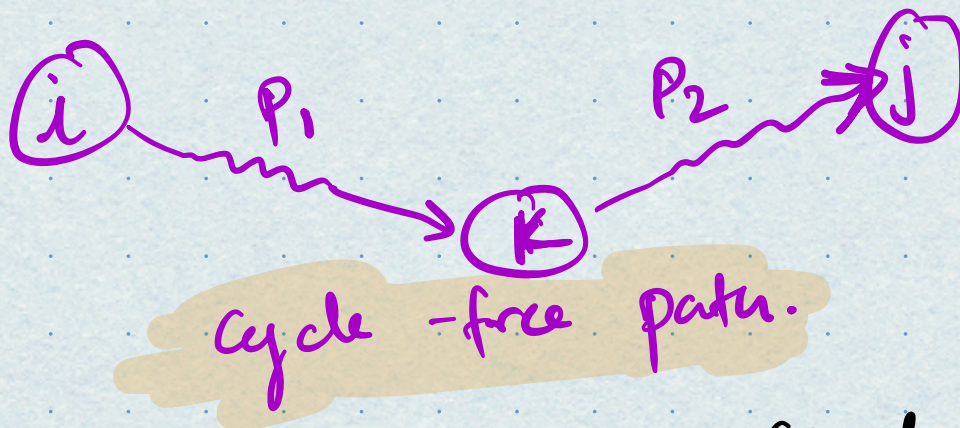
Fix Source $i \in V$,
destination $j \in V$ & prefix length k

$P =$ Shortest (cycle-free) $i \rightsquigarrow j$
path with all internal nodes
in $V^{(k)}$.

Case-1: If k is not
internal to P

$\Rightarrow P$ is the shortest (cycle-free) $i-j$ path with all
internal nodes in $V^{(k-1)}$

Case - 2 : If k is internal
to P



$\Rightarrow P_1$ is the shortest (cycle-free)
 $i-k$ path with all internal
nodes in $V^{(k-1)}$.

P_2 is the shortest (cycle-free)
 $k-j$ path with all internal
nodes in $V^{(k-1)}$.

Let's try edge-Burn



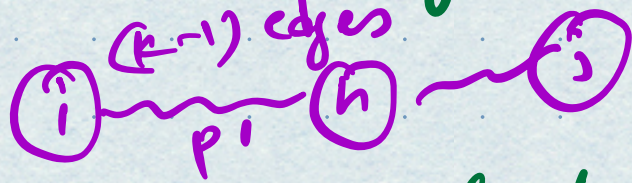
Lemma: Suppose G has no
-ve cycle. Fix a source $i \in V$,
destination $j \in V$, & **edge budget**
 k .

Case -1 :- If P has at
most $(k-1)$ edges

$\Rightarrow P$ is the shortest (cycle-free)
 $i \rightsquigarrow j$ path with at most
 $k-1$ edges.

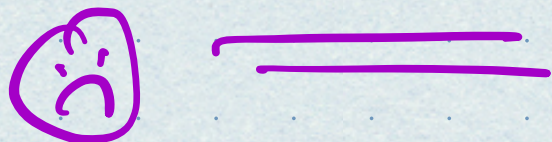
Case-2 :-

If p has k edges



p' is the shortest (cycle free)
 $i \sim h$ path with at most
 $(k-1)$ edges.

Check. - For each i, j
 $O(1 + \text{in-deg}(j))$ time
per update $\Rightarrow O(n^2)$



Back to prefixes

The Recurrence

Set up: Let A be a
3D array (indexed by
 i, j, k)

Intent: -

$A[i, j, k]$ = length of
shortest $i \sim j$ path
with all internal
vertices in $V^{(k)}$

Base Cases :-

What is $A[i, i, 0]$

Three Cases

- $i = j \rightarrow 0$ (empty path)

- $(i, j) \in E$? L_{ij}

- $i \neq j$ and $(i, j) \in E$?

$+\infty$

(No path)

Recurrence

$\forall i \in V, j \in V$ and
 $k \leq |V|$.

$$A[i, j, k] =$$

$$\min \left\{ \begin{array}{l} A[i, j, k-1], \\ A[i, k, k-1] \\ + A[k, j, k-1] \end{array} \right\}$$

Recurrence

for $k = 1$ to n

for $i \in V$

for $j \in V$

$$A[i, j, k] =$$

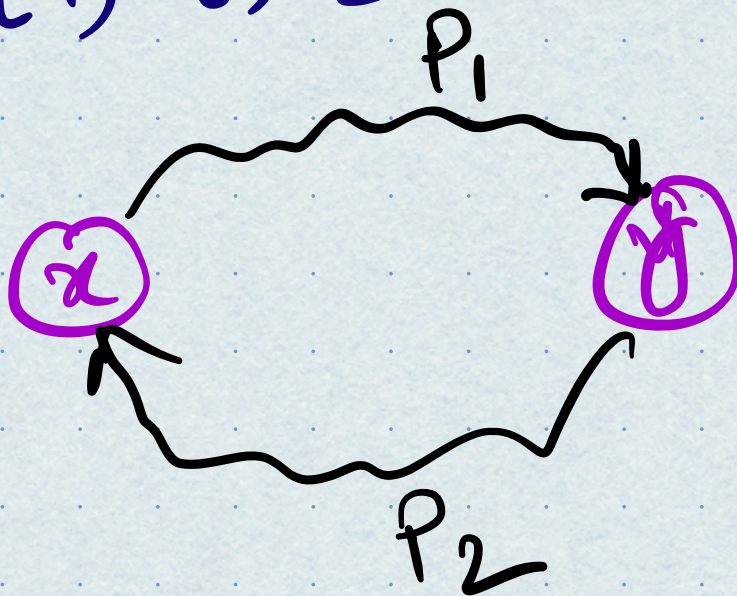
$$\min \left\{ \begin{array}{l} A[i, j, k-1], \\ A[i, k, k-1] \\ + A[k, j, k-1] \end{array} \right\}$$

Run time — $O(1)$

Per subproblem. $O(n^3)$
total

What about -ve
Cycles

Then, at least one
"diagonal" entry
 $A[i, j, n]$ is negative.



$$A[x, x, y] = \min \begin{cases} A[x, x, y-1] \\ A[x, y, y-1] \\ + A[y, x, y-1] \end{cases}$$

< wt. of the cycle < 0

Reconstruction
of shortest paths.

$B[i, j] = \max$ label, say l
of an internal node
on a shortest $i \rightsquigarrow j$
path, $\forall i, j \in V$ (how?)

Recurse on $B[i, l]$
& $B[l, j]$